HSC MATHEMATICS STANDARD 2 · A3 PRACTICE BOOKLET

The questions the state found hardest

50 past HSC questions, ranked from the lowest state mean mark upwards. Every page gives you the original question, the state mean mark, and room to work. Worked solutions with markers' comments start on page 46.

50

Exams covered: 2009–2020 · 8 questions from the current Mathematics Standard 2 exams (2019 onward), 42 from the earlier General Mathematics papers on the same content.
Lowest state mean mark in the set: 1% · highest: 50%
Compiled 6 October 2026 from the SmarterMaths revision worksheets in the shared Year 12 Drive, which quote NESA's state mean mark for each question.

How this booklet was built, and how to use it

Where the percentages come from. After every HSC, NESA records the mean mark the state achieved on each question. The SmarterMaths revision worksheets saved in the shared Year 12 Drive print that figure beside each past-paper question as *“Mean mark xx%”* (the state's average mark as a percentage of the marks available). For a multiple-choice question that is simply the percentage of the state that chose the correct answer. For a multi-part question SmarterMaths sometimes reports the figure only for the hardest part, and sometimes for each part; the booklet shows whatever was reported and ranks each question by its lowest reported figure.

What is in the set. Every Mathematics Standard 2 / General Mathematics HSC question in those worksheets that carries a reported state mean mark, with duplicates removed: 50 questions from the 2009–2020 exams. SmarterMaths only annotates questions the state found hard (roughly 50% or below), so the set is already the hard end of the exam. The worksheets available in the Drive cover statistics (data analysis, bivariate data, the normal distribution), probability, financial mathematics (investments, loans, annuities) and some algebra. They do not cover networks, measurement or trigonometry, so no questions from those topics appear here. Questions from before 2019 come from the General Mathematics / Mathematics General 2 papers; they are included only where SmarterMaths maps them onto the current Standard 2 syllabus.

How to use it.

- Work in rank order if you want the hardest material first, or use the contents page to pick a topic.
- Print on A3 (portrait). Each extended-response question has its own page with working space; multiple-choice questions share a page.
- Do the question under exam conditions before you look at the solution. Allow about 1.5 minutes per mark.
- The colour of the percentage tile flags how badly the state went: red is below 20%, orange is 20–34%, yellow is 35% and above.
- Worked solutions (with SmarterMaths' markers' comments) are collected at the back from page 46, in the same rank order.

Honest caveats. The ranking is by the lowest reported mean mark, so a 1-mark part at 13% sits above a whole 5-mark question at 28%. Two large SmarterMaths worksheets in the Drive (“Displaying Data” and “Summary Statistics”) could not be opened for this build, so a handful of hard data-display questions may be missing. Question images are reproduced from the worksheets (copyright NESA for the exam questions and SmarterMaths for the solutions) for personal study only.

Contents, ranked by state mean mark (lowest first)

Rank	Exam	Question	Syllabus topic	Marks	State mean	By part	Page	Solution
1	2013 HSC General Maths	Q27c	Statistics S1 - Data analysis	3	**1%**	(i) 13% · (ii) 1%	5	46
2	2011 HSC General Maths	Q25a	Data analysis (now Statistics S1)	4	**7%**	(iii) 7% · (iv) 18%	6	46
3	2017 HSC Maths General 2	Q29d	Statistics S5 - The normal distribution	4	**13%**	(i) 50% · (ii) 34% · (iii) 13%	7	46
4	2019 HSC Maths Standard 2	Q15 MC	Statistics S5 - The normal distribution	1	**17%**		8	46
5	2019 HSC Maths Standard 2	Q10 MC	Statistics S1 - Data analysis	1	**18%**		8	46
6	2009 HSC General Maths	Q28b	Statistics S4 - Bivariate data analysis	3	**18%**	(i) 48% · (ii) 18%	9	46
7	2013 HSC General Maths	Q29b	Statistics S5 - The normal distribution	4	**21%**	(i) 39% · (ii) 21%	10	46
8	2017 HSC Maths General 2	Q27c	Financial F5 - Annuities	2	**22%**	part (ii)	11	46
9	2017 HSC Maths General 2	Q30d	Algebra (now A2 Linear relationships)	3	**23%**	part (ii)	12	46
10	2013 HSC General Maths	Q20 MC	Statistics S5 - The normal distribution	1	**24%**		13	47
11	2012 HSC General Maths	Q9 MC	Financial F4 - Investments and loans	1	**24%**		13	47
12	2016 HSC Maths General 2	Q7 MC	Data analysis (now Statistics S1)	1	**26%**		14	47
13	2019 HSC Maths Standard 2	Q42	Financial F5 - Annuities	3	**27%**		15	47
14	2018 HSC Maths General 2	Q26e	Statistics S1 - Data analysis	2	**27%**		16	47
15	2019 HSC Maths Standard 2	Q39	Statistics S1 - Data analysis	5	**28%**		17	47
16	2011 HSC General Maths	Q23c	Financial F4 - Investments and loans	2	**28%**		18	47
17	2009 HSC General Maths	Q21 MC	Statistics S1 - Data analysis	1	**28%**		14	47
18	2009 HSC General Maths	Q27a	Financial F5 - Annuities	5	**31%**	(ii) 45% · (iii) 31%	19	48
19	2020 HSC Maths Standard 2	Q7 MC	Statistics S1 - Data analysis	1	**32%**		20	48
20	2016 HSC Maths General 2	Q30d	Statistics S5 - The normal distribution	4	**32%**	part (ii)	21	48
21	2018 HSC Maths General 2	Q19 MC	Financial F4 - Investments and loans	1	**33%**		20	48
22	2016 HSC Maths General 2	Q29e	Algebra A2 - Linear relationships	3	**33%**	part (ii)	22	48
23	2015 HSC Maths General 2	Q30c	Financial F5 - Annuities	3	**33%**	(i) 48% · (ii) 33%	23	48
24	2016 HSC Maths General 2	Q23 MC	Statistics S2 - Relative frequency and probability	1	**34%**		24	48
25	2011 HSC General Maths	Q25d	Statistics S1 - Data analysis	2	**34%**	part (ii)	25	48

Contents, ranked by state mean mark (lowest first) (continued)

Rank	Exam	Question	Syllabus topic	Marks	State mean	By part	Page	Solution
26	2013 HSC General Maths	Q28b	Bivariate data (now Statistics S4)	6	**38%**	part (ii)	26	49
27	2015 HSC Maths General 2	Q19 MC	Bivariate data (now Statistics S4)	1	**39%**		24	49
28	2019 HSC Maths Standard 2	Q23	Statistics S4 - Bivariate data analysis	3	**40%**	part (a)	27	49
29	2018 HSC Maths General 2	Q26a	Probability (now Statistics S2)	1	**40%**		28	49
30	2016 HSC Maths General 2	Q28d	Financial F5 - Annuities	2	**40%**		29	49
31	2013 HSC General Maths	Q23 MC	Financial F5 - Annuities	1	**41%**		30	49
32	2009 HSC General Maths	Q25d	Statistics S5 - The normal distribution	4	**41%**	part (ii)	31	49
33	2019 HSC Maths Standard 2	Q34	Algebra A2 - Linear relationships	3	**42%**	part (a)	32	50
34	2011 HSC General Maths	Q27d	Financial F5 - Annuities	4	**42%**		33	50
35	2011 HSC General Maths	Q27c	Statistics S5 - The normal distribution	3	**42%**	part (ii)	34	50
36	2010 HSC General Maths	Q24c	Statistics S5 - The normal distribution	4	**42%**	part (ii)	35	50
37	2016 HSC Maths General 2	Q21 MC	Statistics S1 - Data analysis	1	**43%**		30	50
38	2014 HSC Maths General 2	Q21 MC	Financial F5 - Annuities	1	**43%**		36	50
39	2012 HSC General Maths	Q11 MC	Statistics S4 - Bivariate data analysis	1	**43%**		36	50
40	2019 HSC Maths Standard 2	Q38	Statistics S5 - The normal distribution	2	**44%**		37	50
41	2015 HSC Maths General 2	Q28b	Statistics S5 - The normal distribution	2	**44%**		38	51
42	2013 HSC General Maths	Q26e	Financial F4 - Investments and loans	2	**44%**		39	51
43	2012 HSC General Maths	Q29b	Statistics S5 - The normal distribution	2	**44%**		40	51
44	2014 HSC Maths General 2	Q24 MC	Statistics S5 - The normal distribution	1	**46%**		41	51
45	2018 HSC Maths General 2	Q23 MC	Statistics S5 - The normal distribution	1	**47%**		41	51
46	2017 HSC Maths General 2	Q27a	Statistics S1 - Data analysis	2	**47%**	part (ii)	42	51
47	2017 HSC Maths General 2	Q29c	Probability (now Statistics S2)	4	**47%**	part (ii)	43	51
48	2014 HSC Maths General 2	Q30a	Financial F4 - Investments and loans	3	**49%**		44	51
49	2015 HSC Maths General 2	Q20 MC	Statistics S5 - The normal distribution	1	**50%**		45	52
50	2011 HSC General Maths	Q20 MC	Algebra A4 - Types of relationships	1	**50%**		45	52

#1

2013 HSC General Mathematics · Question 27c

Statistics S1 - Data analysis · 3 marks

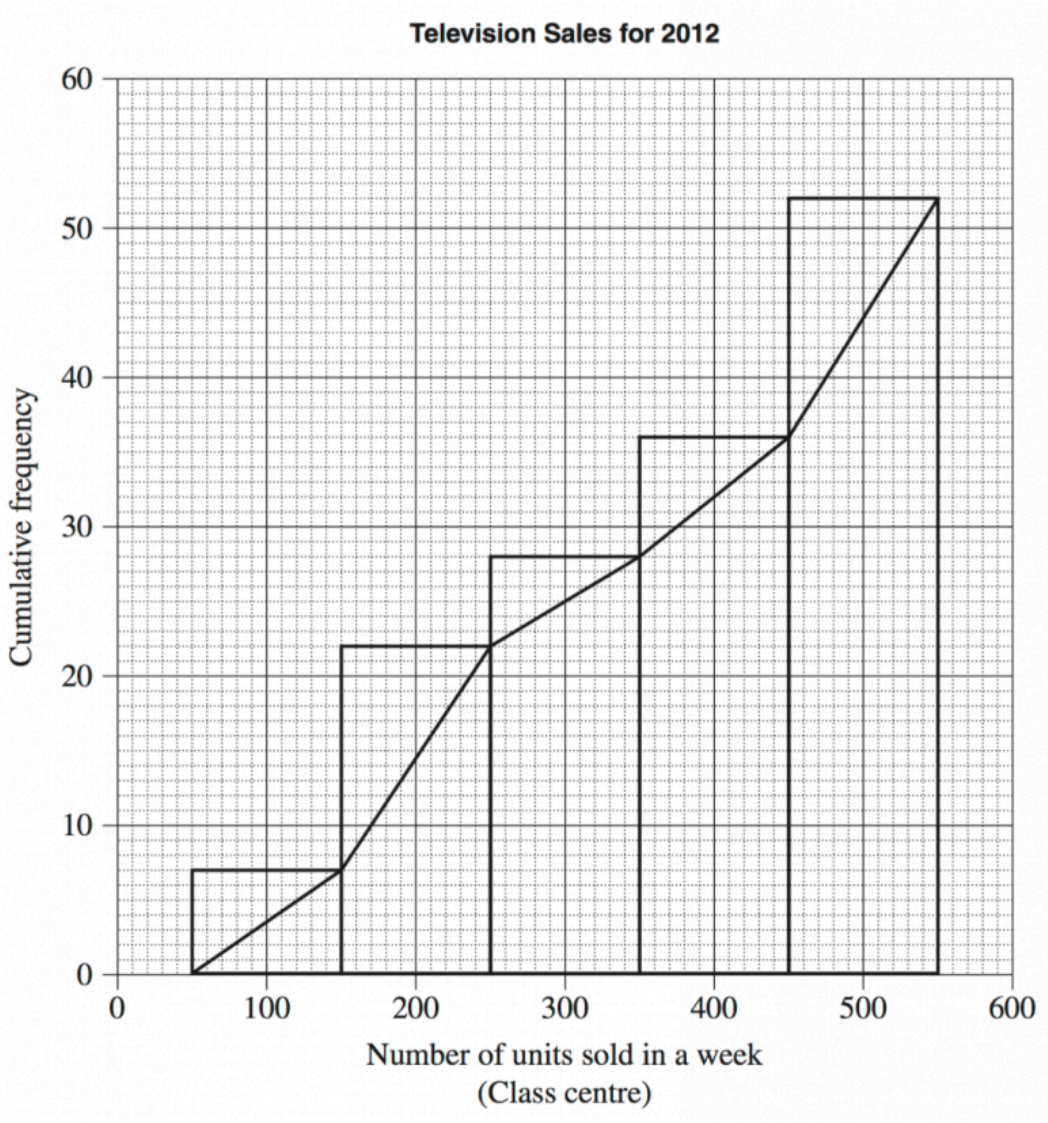
1%

state mean mark, part (ii)

State mean mark by part: (i) 13% · (ii) 1%

A retailer has collected data on the number of televisions that he sold each week in 2012.

He grouped the data into classes and displayed the data using a cumulative frequency histogram and polygon (ogive).



i. Use the cumulative frequency polygon to determine the interquartile range. *(2 marks)*

.....

.....

.....

.....

ii. Oscar said that the retailer sold 300 televisions in 6 of the weeks in 2012.

Is he correct? Give a reason for your answer. *(1 mark)*

WORKING

#2

2011 HSC General Mathematics · Question 25a

Data analysis (now Statistics S1) · 4 marks

7%

state mean mark, part (iii)

State mean mark by part: (iii) 7% · (iv) 18%

A study on the mobile phone usage of NSW high school students is to be conducted.

Data is to be gathered using a questionnaire.

The questionnaire begins with the three questions shown.

Q1: Do you own a mobile phone?

Yes ☐ No ☐

Q2: Which phone company do you use?

.....

Q3: Do you use pre-paid or a plan?

Pre-paid ☐ Plan ☐

(i) Classify the type of data that will be collected in Q2 of the questionnaire. ***(1 mark)***

(ii) Write a suitable question for this questionnaire that would provide quantitative data. ***(1 mark)***

(iii) An initial study is to be conducted using a stratified sample.

Describe a method that could be used to obtain a representative stratified sample. ***(1 mark)***

(iv) Who should be surveyed if it is decided to use a census for the study? ***(1 mark)***

WORKING

#3

2017 HSC Mathematics General 2 · Question 29d

Statistics S5 - The normal distribution · 4 marks

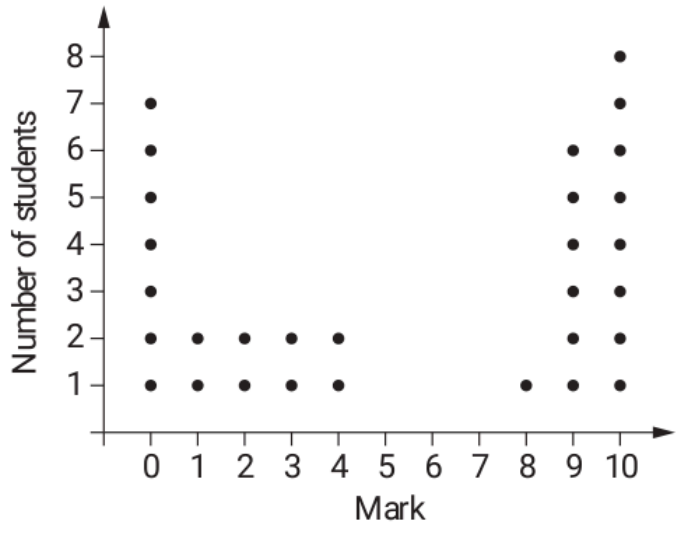
13%

state mean mark, part (iii)

State mean mark by part: (i) 50% · (ii) 34% · (iii) 13%

All the students in a class of 30 did a test.

The marks, out of 10, are shown in the dot plot.



i. Find the median test mark. ***(1 mark)***

ii. The mean test mark is 5.4. The standard deviation of the test marks is 4.22.

Using the dot plot, calculate the percentage of the marks which lie within one standard deviation of the mean. ***(2 marks)***

iii. A student states that for any data set, 68% of the scores should lie within one standard deviation of the mean. With reference to the dot plot, explain why the student's statement is NOT relevant in this context. ***(1 mark)***

WORKING

#4 2019 HSC Mathematics Standard 2 · Question 15

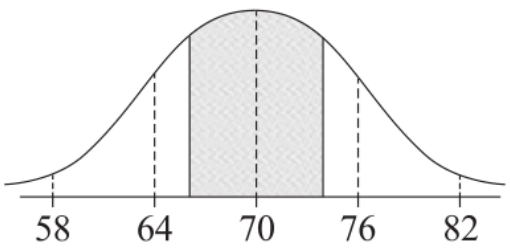
Statistics S5 - The normal distribution · Multiple choice · 1 mark

17%
state mean mark

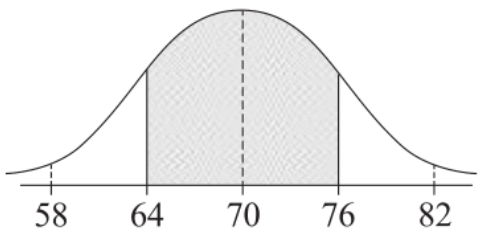
The scores on an examination are normally distributed with a mean of 70 and a standard deviation of 6. Michael received a score on the examination between the lower quartile and the upper quartile of the scores.

Which shaded region most accurately represents where Michael's score lies?

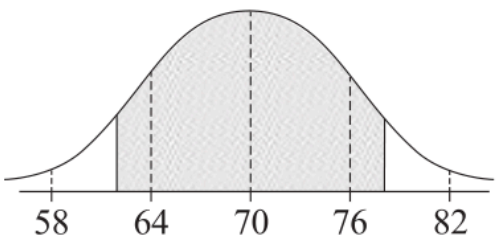
A.



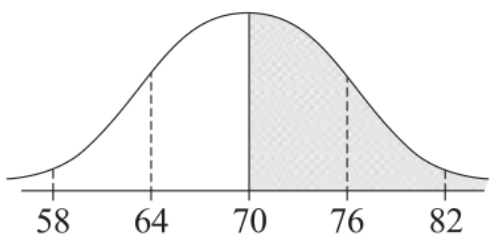
B.



C.



D.



Answer: (A) (B) (C) (D)

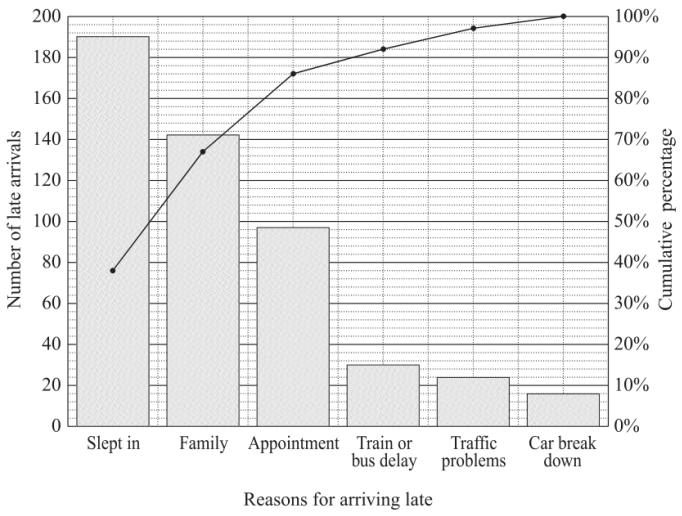
WORKING

#5 2019 HSC Mathematics Standard 2 · Question 10

Statistics S1 - Data analysis · Multiple choice · 1 mark

18%
state mean mark

A school collected data related to the reasons given by students for arriving late. The Pareto chart shows the data collected.



What percentage of students gave the reason 'Train or bus delay'?

A. 6%
B. 15%
C. 30%
D. 92%

Answer: (A) (B) (C) (D)

WORKING

#6

2009 HSC General Mathematics · Question 28b

Statistics S4 - Bivariate data analysis · 3 marks

18%

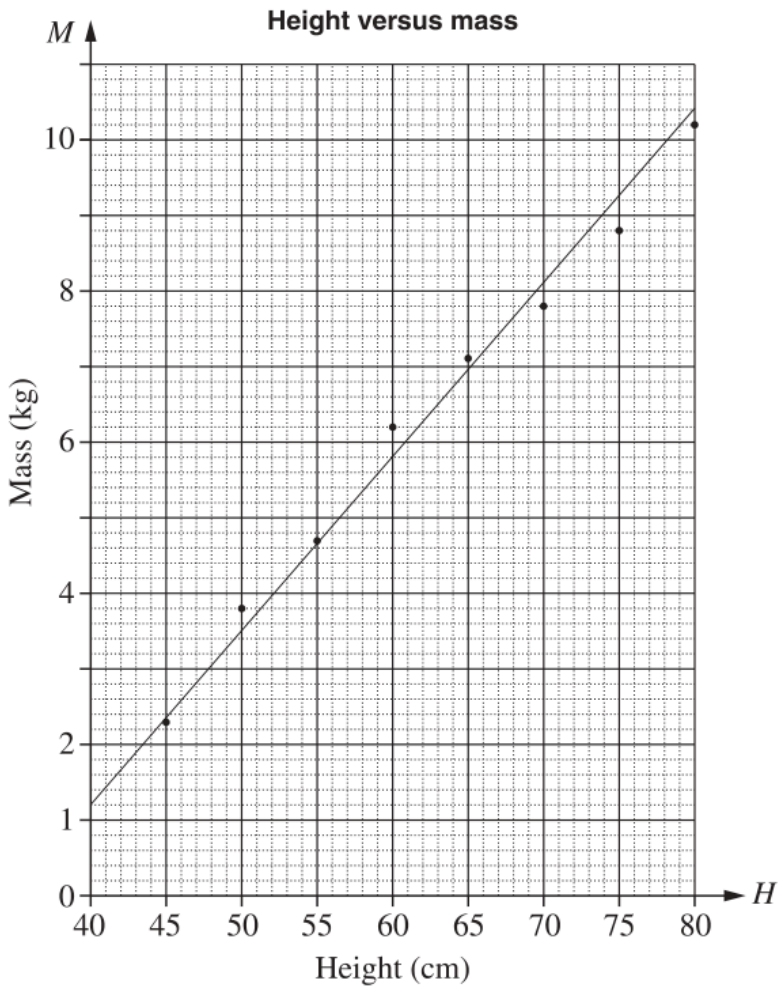
state mean mark, part (ii)

State mean mark by part: (i) 48% · (ii) 18%

The height and mass of a child are measured and recorded over its first two years.

Height (cm), H	45	50	55	60	65	70	75	80
Mass (kg), M	2.3	3.8	4.7	6.2	7.1	7.8	8.8	10.2

This information is displayed in a scatter graph.



i. Describe the correlation between the height and mass of this child, as shown in the graph. *(1 mark)*

.....

.....

ii. A line of best fit has been drawn on the graph.

Find the equation of this line. *(2 marks)*

WORKING

#7

2013 HSC General Mathematics · Question 29b

Statistics S5 - The normal distribution · 4 marks

21%

state mean mark, part (ii)

State mean mark by part: (i) 39% · (ii) 21%

Ali's class sits two Geography tests. The results of her class on the first Geography test are shown.

$$58,\ 74,\ 65,\ 66,\ 73,\ 71,\ 72,\ 74,\ 62,\ 70$$

The mean was 68.5 for the first test.

(i) Calculate the standard deviation for the first test. Give your answer correct to one decimal place. ***(1 mark)***

(ii) On the second Geography test, the mean for the class was 74.4 and the standard deviation was 12.4.
Ali scored 62 on the first test. Calculate the mark that she needed to obtain in the second test to ensure that her performance relative to the class was maintained. ***(3 marks)***

WORKING

#8

2017 HSC Mathematics General 2 · Question 27c

Financial F5 - Annuities · 2 marks

22%

state mean mark, part (ii)

A table of future value interest factors for an annuity of $1 is shown.

Table of future value interest factors					
Period	*Interest rate per period*				
	1%	2%	3%	4%	5%
3	3.0301	3.0604	3.0909	3.1216	3.1525
4	4.0604	4.1216	4.1836	4.2465	4.3101
5	5.1010	5.2040	5.3091	5.4163	5.5256
6	6.1520	6.3081	6.4684	6.6330	6.8019

An annuity involves contributions of $12 000 per annum for 5 years. The interest rate is 4% per annum, compounded annually.

(i) Calculate the future value of this annuity. ***(1 mark)***

(ii) Calculate the interest earned on this annuity. ***(1 mark)***

WORKING

#9

2017 HSC Mathematics General 2 · Question 30d

Algebra (now A2 Linear relationships) · 3 marks

23%

state mean mark, part (ii)

In an investigation, students used different numbers of identical small solar panels to power model cars. The cars were then tested and their speed measured in km/h. The results are summarised in the table.

	Mean	*Standard deviation*
Number of solar panels (x)	2.9	0.8
Speed (y)	8.2	2

The equation of the least-squares line of best fit, relating the speed and the number of solar panels, has been calculated to be

$$y = 2.125x + 2.0375$$

(i) What would be the speed of a car powered by 5 solar panels, based on this equation? ***(1 mark)***

(ii) Calculate the correlation coefficient, r, between the number of solar panels and the speed of a car. ***(2 marks)***

WORKING

#10 2013 HSC General Mathematics · Question 20

Statistics S5 - The normal distribution · Multiple choice · 1 mark

24%
state mean mark

There are 60 000 students sitting a state-wide examination. If the results form a normal distribution, how many students would be expected to score a result between 1 and 2 standard deviations above the mean?

You may assume for normally distributed data that:

- 68% of scores have z-scores between -1 and 1
- 95% of scores have z-scores between -2 and 2
- 99.7% of scores have z-scores between -3 and 3.

A. 8100

B. 16 200

C. 20 400

D. 28 500

Answer: (A) (B) (C) (D)

WORKING

#11 2012 HSC General Mathematics · Question 9

Financial F4 - Investments and loans · Multiple choice · 1 mark

24%
state mean mark

Tracy invests some money for 2 years at 4% per annum, compounded quarterly.

Compounded values of $1

Period	Interest rate per period				
	1%	*2%*	*3%*	*4%*	*5%*
1	1.010	1.020	1.030	1.040	1.050
2	1.020	1.040	1.061	1.082	1.103
3	1.030	1.061	1.093	1.125	1.158
4	1.041	1.082	1.126	1.170	1.216
5	1.051	1.104	1.159	1.217	1.276
6	1.062	1.126	1.194	1.265	1.340
7	1.072	1.149	1.230	1.316	1.407
8	1.083	1.172	1.267	1.369	1.477

Which figure from the table should Tracy use to calculate the value of her investment at the end of 2 years?

(A) 1.020

(B) 1.082

(C) 1.083

(D) 1.369

Answer: (A) (B) (C) (D)

WORKING

#12 2016 HSC Mathematics General 2 · Question 7

Data analysis (now Statistics S1) · Multiple choice · 1 mark

26% state mean mark

Which set of data is classified as categorical and nominal?

(A) blue, green, yellow

(B) small, medium, large

(C) 5.2 cm, 6 cm, 7.21 cm

(D) 4 people, 5 people, 9 people

Answer: (A) (B) (C) (D)

WORKING

#17 2009 HSC General Mathematics · Question 21

Statistics S1 - Data analysis · Multiple choice · 1 mark

28% state mean mark

The mean of a set of ten scores is 14. Another two scores are included and the new mean is 16.

What is the mean of the two additional scores?

A. 4

B. 16

C. 18

D. 26

Answer: (A) (B) (C) (D)

WORKING

#13

2019 HSC Mathematics Standard 2 · Question 42

Financial F5 - Annuities · 3 marks

27%
state mean mark

The table shows the future values of an annuity of $1 for different interest rates for 4, 5 and 6 years. The contributions are made at the end of each year.

Future value of an annuity of $1

Years	*Interest rate per annum*			
	1%	*2%*	*3%*	*4%*
4	4.060	4.122	4.184	4.246
5	5.101	5.204	5.309	5.416
6	6.152	6.308	6.468	6.633

An annuity account is opened and contributions of $2000 are made at the end of each year for 7 years.

For the first 6 years, the interest rate is 4% per annum, compounding annually.

For the 7th year, the interest rate increases to 5% per annum, compounding annually.

Calculate the amount in the account immediately after the 7th contribution is made. ***(3 marks)***

WORKING

#14

2018 HSC Mathematics General 2 · Question 26e

Statistics S1 - Data analysis · 2 marks

27%
state mean mark

A cumulative frequency table for a data set is shown.

Score	*Cumulative frequency*
1	5
2	9
3	16
4	20
5	34
6	42

What is the interquartile range of this data set? ***(2 marks)***

WORKING

#15

2019 HSC Mathematics Standard 2 · Question 39

Statistics S1 - Data analysis · 5 marks

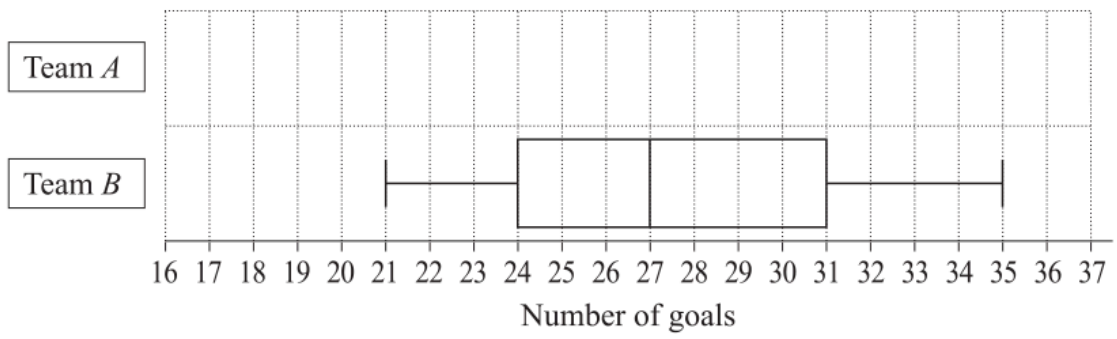
28%
state mean mark

Two netball teams, Team A and Team B, each played 15 games in a tournament. For each team, the number of goals scored in each game was recorded.

The frequency table shows the data for Team A.

Number of goals	*Frequency*
19	1
20	0
21	1
22	1
23	1
24	3
25	0
26	4
27	3
28	1

The data for Team B was analysed to create the box-plot shown.



Compare the distributions of the number of goals scored by the two teams. Support your answer with the construction of a box-plot for the data for Team A. ***(5 marks)***

WORKING

#16

2011 HSC General Mathematics · Question 23c

Financial F4 - Investments and loans · 2 marks

28%
state mean mark

An amount of $5000 is invested at 10% per annum, compounded six-monthly.

Compounded values of $1

Period	Interest rate per period				
	1%	*5%*	*10%*	*15%*	*20%*
1	1.010	1.050	1.100	1.150	1.200
2	1.020	1.103	1.210	1.323	1.440
3	1.030	1.158	1.331	1.521	1.728
4	1.041	1.216	1.464	1.750	2.074
5	1.051	1.276	1.611	2.011	2.488
6	1.062	1.340	1.772	2.313	2.986

Use the table to find the value of this investment at the end of three years. ***(2 marks)***

WORKING

#18

2009 HSC General Mathematics · Question 27a

Financial F5 - Annuities · 5 marks

31%

state mean mark, part (iii)

State mean mark by part: (ii) 45% · (iii) 31%

The table shows the future value of a $1 annuity at different interest rates over different numbers of time periods.

Future values of a $1 annuity

Time Period	*Interest rate*				
	1%	*2%*	*3%*	*4%*	*5%*
1	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0100	2.0200	2.0300	2.0400	2.0500
3	3.0301	3.0604	3.0909	3.1216	3.1525
4	4.0604	4.1216	4.1836	4.2465	4.3101
5	5.1010	5.2040	5.3091	5.4163	5.5256
6	6.1520	6.3081	6.4684	6.6330	6.8019
7	7.2135	7.4343	7.6625	7.8983	8.1420
8	8.2857	8.5830	8.8923	9.2142	9.5491

(i) What would be the future value of a $5000 per year annuity at 3% per annum for 6 years, with interest compounding yearly? ***(1 mark)***

(ii) What is the value of an annuity that would provide a future value of $407 100 after 7 years at 5% per annum compound interest? ***(1 mark)***

(iii) An annuity of $1000 per quarter is invested at 4% per annum, compounded quarterly for 2 years. What will be the amount of interest earned? ***(3 marks)***

WORKING

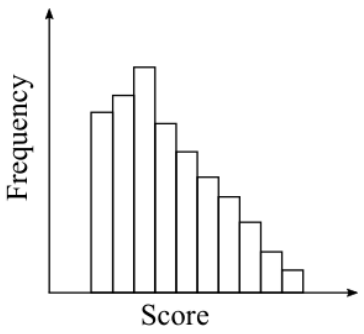
#19 2020 HSC Mathematics Standard 2 · Question 7

Statistics S1 - Data analysis · Multiple choice · 1 mark

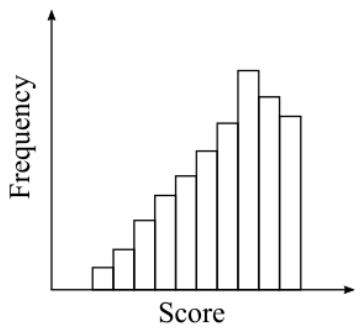
32%
state mean mark

Which histogram best represents a dataset that is positively skewed?

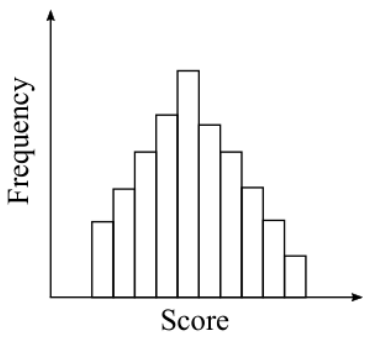
A.



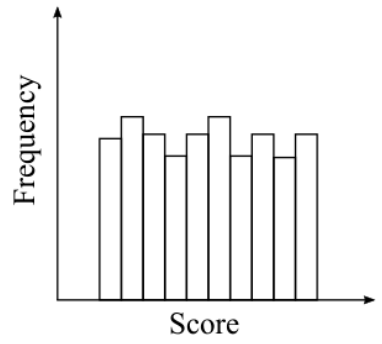
B.



C.



D.



Answer: (A) (B) (C) (D)

WORKING

#21 2018 HSC Mathematics General 2 · Question 19

Financial F4 - Investments and loans · Multiple choice · 1 mark

33%
state mean mark

The table shows the compounded values of $1 at different interest rates over different periods.

Compounded values of $1

Number of periods	*Interest rate per period*				
	1%	2%	3%	4%	5%
2	1.0201	1.0404	1.0609	1.0816	1.1025
4	1.0406	1.0824	1.1255	1.1699	1.2155
6	1.0615	1.1262	1.1941	1.2653	1.3401
8	1.0829	1.1717	1.2668	1.3686	1.4775
10	1.1046	1.2190	1.3439	1.4802	1.6289
12	1.1268	1.2682	1.4258	1.6010	1.7959

Amy hopes to have $21 000 in 2 years to buy a car. She opens an account today which pays interest of 4% pa, compounded quarterly.

Using the table, which expression calculates the minimum single sum that Amy needs to invest today to ensure she reaches her savings goal?

A. 21 000 × 1.0816

B. 21 000 ÷ 1.0816

C. 21 000 × 1.0829

D. 21 000 ÷ 1.0829

Answer: (A) (B) (C) (D)

WORKING

#20

2016 HSC Mathematics General 2 · Question 30d

Statistics S5 - The normal distribution · 4 marks

32%

state mean mark, part (ii)

The formula to calculate z-scores can be rearranged to give

$$\mu = x - \sigma z$$

where μ is the mean

x is the score

σ is the standard deviation

z is the z-score

i. In an examination, Aaron achieved a score of 88, which corresponds to a z-score of 2.4.
Substitute these values into the rearranged formula above to form an equation. ***(1 mark)***

ii. In the same examination, Brock achieved a score of 52, which corresponds to a z-score of -1.2.
Using this information, form another equation and solve it simultaneously with the equation from part (i) to find the values of μ and σ. ***(3 marks)***

WORKING

#22

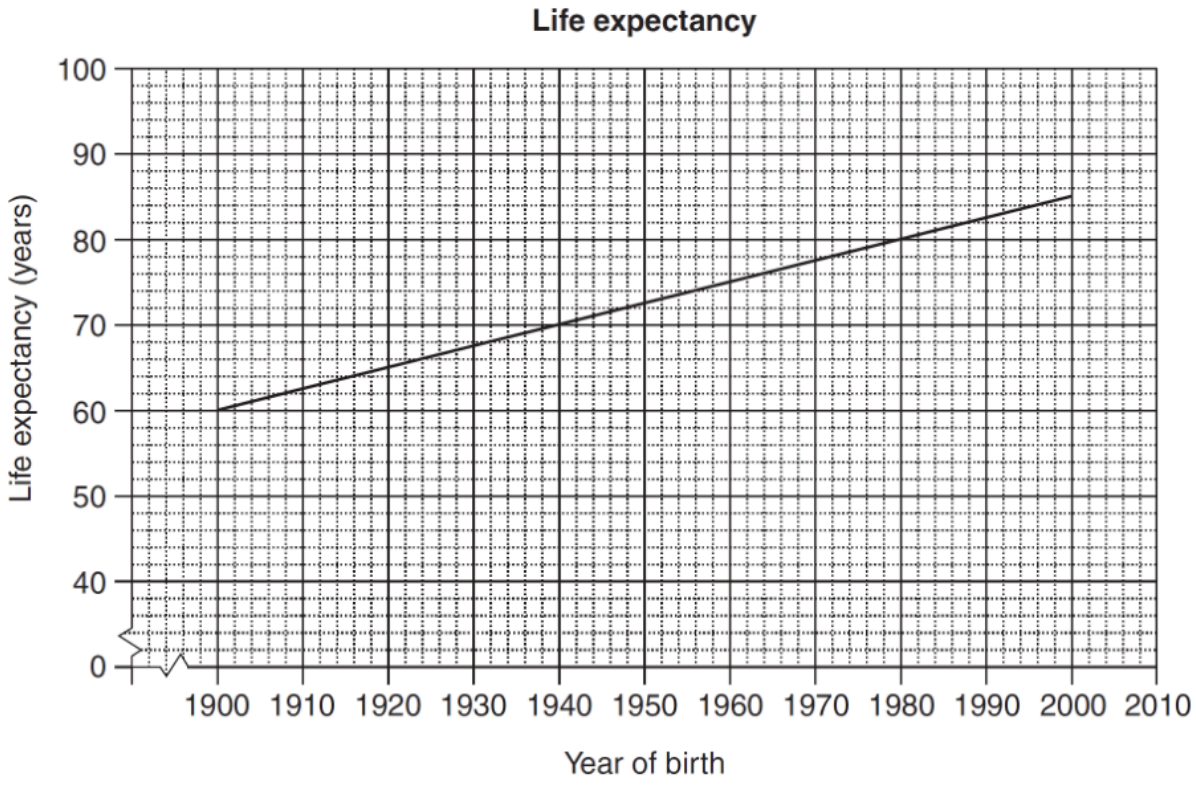
2016 HSC Mathematics General 2 · Question 29e

Algebra A2 - Linear relationships · 3 marks

33%

state mean mark, part (ii)

The graph shows the life expectancy of people born between 1900 and 2000.



(i) According to the graph, what is the life expectancy of a person born in 1932? ***(1 mark)***

(ii) With reference to the value of the gradient, explain the meaning of the gradient in this context. ***(2 marks)***

WORKING

#23

2015 HSC Mathematics General 2 · Question 30c

Financial F5 - Annuities · 3 marks

33%

state mean mark, part (ii)

State mean mark by part: (i) 48% · (ii) 33%

The table gives the present value interest factors for an annuity of \$1 per period, for various interest rates (r) and numbers of periods (N).

Table of present value interest factors

N \ r	Interest rate per period (as a decimal)				
	0.0075	0.0080	0.0085	0.0090	0.0095
70	54.30462	53.43960	52.59397	51.76724	50.95891
71	54.89293	54.00754	53.14226	52.29657	51.46995
72	55.47685	54.57097	53.68593	52.82118	51.97618
73	56.05643	55.12993	54.22502	53.34111	52.47764
74	56.63169	55.68446	54.75957	53.85641	52.97438

i. Oscar plans to invest \$200 each month for 74 months. His investment will earn interest at the rate of 0.0080 (as a decimal) per month.

Use the information in the table to calculate the present value of this annuity. ***(1 mark)***

ii. Lucy is using the same table to calculate the loan repayment for her car loan. Her loan is $\$21\,500$ and will be repaid in equal monthly repayments over 6 years. The interest rate on her loan is 10.8% per annum.

Calculate the amount of each monthly repayment, correct to the nearest dollar. ***(2 marks)***

WORKING

#24 2016 HSC Mathematics General 2 · Question 23

Statistics S2 - Relative frequency and probability · Multiple choice · 1 mark

34%
state mean mark

A group of 485 people was surveyed. The people were asked whether or not they smoke. The results are recorded in the table.

	Smokers	*Non-smokers*	*Total*
Male	88	176	264
Female	68	153	221
	156	329	485

A person is selected at random from the group.

What is the approximate probability that the person selected is a smoker OR is male?

A. 33%

B. 18%

C. 68%

D. 87%

Answer: (A) (B) (C) (D)

WORKING

#27 2015 HSC Mathematics General 2 · Question 19

Bivariate data (now Statistics S4) · Multiple choice · 1 mark

39%
state mean mark

The table shows the life expectancy (expected remaining years of life) for females at selected ages in the given periods of time.

Life expectancy for females

Period of time	*Remaining years of life*			
	at age 0	at age 25	at age 45	at age 65
1965–1967	74.2	51.2	32.3	15.7
1975–1977	76.6	53.1	34.0	17.1
1985–1987	79.2	55.4	36.1	18.6
1995–1997	81.3	57.1	37.7	19.8

In 1975, a 45-year-old female used the information in the table to calculate the age to which she was expected to live. Twenty years later she recalculated the age to which she was expected to live.

What is the difference between the two ages she calculated?

(A) 2.7 years

(B) 3.1 years

(C) 3.7 years

(D) 5.8 years

Answer: (A) (B) (C) (D)

WORKING

#25

2011 HSC General Mathematics · Question 25d

Statistics S1 - Data analysis · 2 marks

34%

state mean mark, part (ii)

Data was collected from 30 students on the number of text messages they had sent in the previous 24 hours. The set of data collected is displayed.

Male		*Female*
9 9 8 7 6 5 5 4 2 1	0	8 9
1 1 0 0	1	1 1 2 5 6 8 8 8
0	2	0 1 7
	3	4
	4	
	5	
	6	
1	7	

i. What is the outlier for this set of data? ***(1 mark)***

.....

ii. What is the interquartile range of the data collected from the female students? ***(1 mark)***

WORKING

#26

2013 HSC General Mathematics · Question 28b

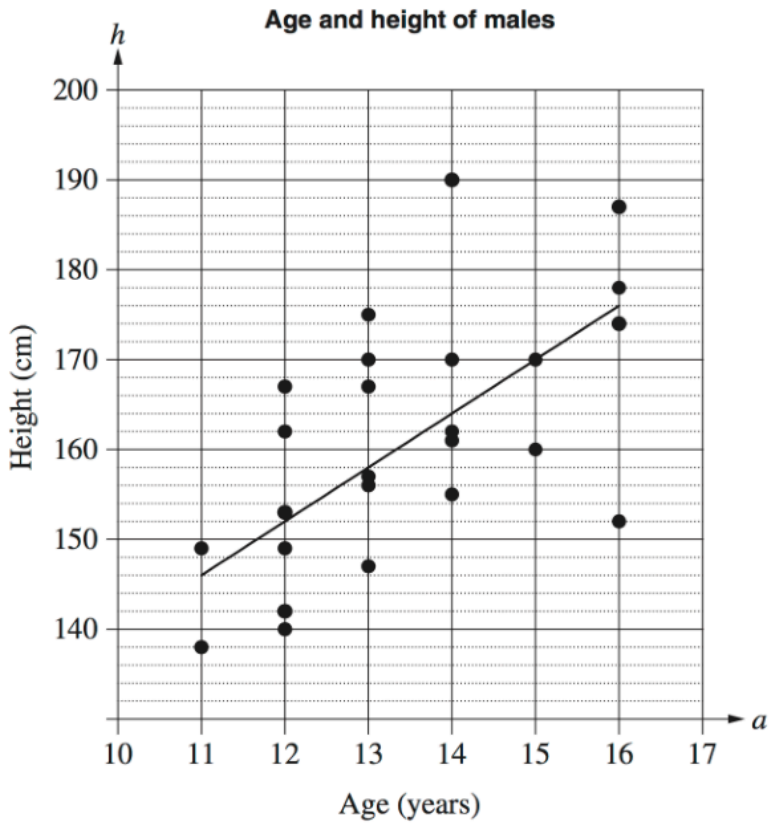
Bivariate data (now Statistics S4) · 6 marks

38%

state mean mark, part (ii)

Ahmed collected data on the age (a) and height (h) of males aged 11 to 16 years.

He created a scatterplot of the data and constructed a line of best fit to model the relationship between the age and height of males.



(i) Determine the gradient of the line of best fit shown on the graph. ***(1 mark)***

(ii) Explain the meaning of the gradient in the context of the data. ***(1 mark)***

(iii) Determine the equation of the line of best fit shown on the graph. ***(2 marks)***

(iv) Use the line of best fit to predict the height of a typical 17-year-old male. ***(1 mark)***

(v) Why would this model not be useful for predicting the height of a typical 45-year-old male? ***(1 mark)***

WORKING

#28

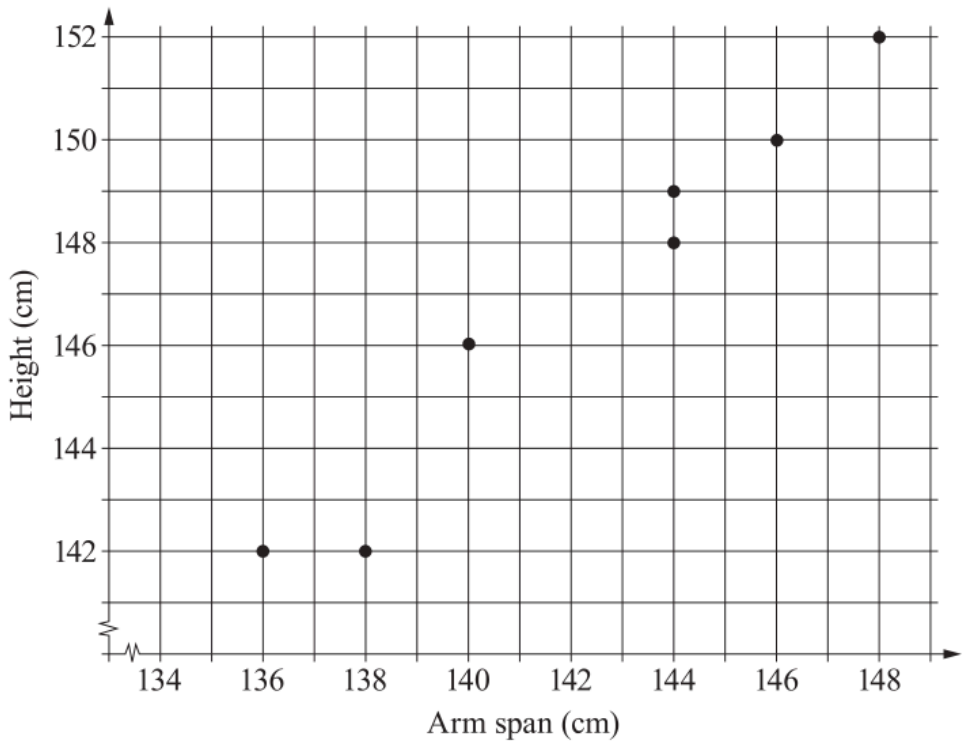
2019 HSC Mathematics Standard 2 · Question 23

Statistics S4 - Bivariate data analysis · 3 marks

40%

state mean mark, part (a)

A set of bivariate data is collected by measuring the height and arm span of seven children. The graph shows a scatterplot of these measurements.



a. Calculate Pearson's correlation coefficient for the data, correct to two decimal places. *(1 mark)*

.....

b. Identify the direction and the strength of the linear association between height and arm span.
(1 mark)

.....

.....

c. The equation of the least-squares regression line is shown.

Height = 0.866 × (arm span) + 23.7

A child has an arm span of 143 cm.

Calculate the predicted height for this child using the equation of the least-squares regression line. *(1 mark)*

WORKING

#29

2018 HSC Mathematics General 2 · Question 26a

Probability (now Statistics S2) · 1 mark

40%
state mean mark

Jeremy rolled a biased 6-sided die a number of times. He recorded the results in a table.

Number	1	2	3	4	5	6
Frequency	23	19	48	20	21	19

What is the relative frequency of rolling a 3? ***(1 mark)***

WORKING

#30

2016 HSC Mathematics General 2 · Question 28d

Financial F5 - Annuities · 2 marks

40%
state mean mark

The table gives the contribution per period for an annuity with a future value of $1 at different interest rates and different periods of time.

Contribution per period for an annuity with a future value of $1

Number of periods	*Interest rate* (% per period)					
	0.25%	0.5%	0.75%	1%	1.25%	1.5%
6	0.1656	0.1646	0.1636	0.1625	0.1615	0.1605
12	0.0822	0.0811	0.0800	0.0788	0.0778	0.0767
18	0.0544	0.0532	0.0521	0.0510	0.0499	0.0488
24	0.0405	0.0393	0.0382	0.0371	0.0360	0.0349
30	0.0321	0.0310	0.0298	0.0287	0.0277	0.0266
36	0.0266	0.0254	0.0243	0.0232	0.0222	0.0212

Margaret needs to save $75 000 over 6 years for a deposit on a new apartment. She makes regular quarterly contributions into an investment account which pays interest at 3% pa.

How much will Margaret need to contribute each quarter to reach her savings goal? ***(2 marks)***

WORKING

#31 2013 HSC General Mathematics · Question 23

Financial F5 - Annuities · Multiple choice · 1 mark

41% state mean mark

Zina opened an account to save for a new car. Six months after opening the account, she made first deposit of $1200 and continued depositing $1200 at the end of each six month period. Interest was paid at 3% per annum, compounded half-yearly.

How much was in Zina's account two years after first opening it?

A. $4909.08

B. $4982.72

C. $5018.16

D. $5094.55

Answer: (A) (B) (C) (D)

WORKING

#37 2016 HSC Mathematics General 2 · Question 21

Statistics S1 - Data analysis · Multiple choice · 1 mark

43% state mean mark

A grouped data frequency table is shown.

Class interval	*Frequency*
1–5	3
6–10	6
11–15	8
16–20	9

What is the mean for this set of data?

A. 6.5

B. 10.5

C. 11.9

D. 12.4

Answer: (A) (B) (C) (D)

WORKING

#32

2009 HSC General Mathematics · Question 25d

Statistics S5 - The normal distribution · 4 marks

41%

state mean mark, part (ii)

In Broken Hill, the maximum temperature for each day has been recorded. The mean of these maximum temperatures during spring is 25.8°C, and their standard deviation is 4.2° C.

(i) What temperature has a z-score of -1? ***(1 mark)***

(ii) What percentage of spring days in Broken Hill would have maximum temperatures between 21.6° C and 38.4°C?
You may assume that these maximum temperatures are normally distributed and that

- 68% of maximum temperatures have z-scores between -1 and 1
- 95% of maximum temperatures have z-scores between -2 and 2
- 99.7% of maximum temperatures have z-scores between -3 and 3. ***(3 marks)***

WORKING

#33

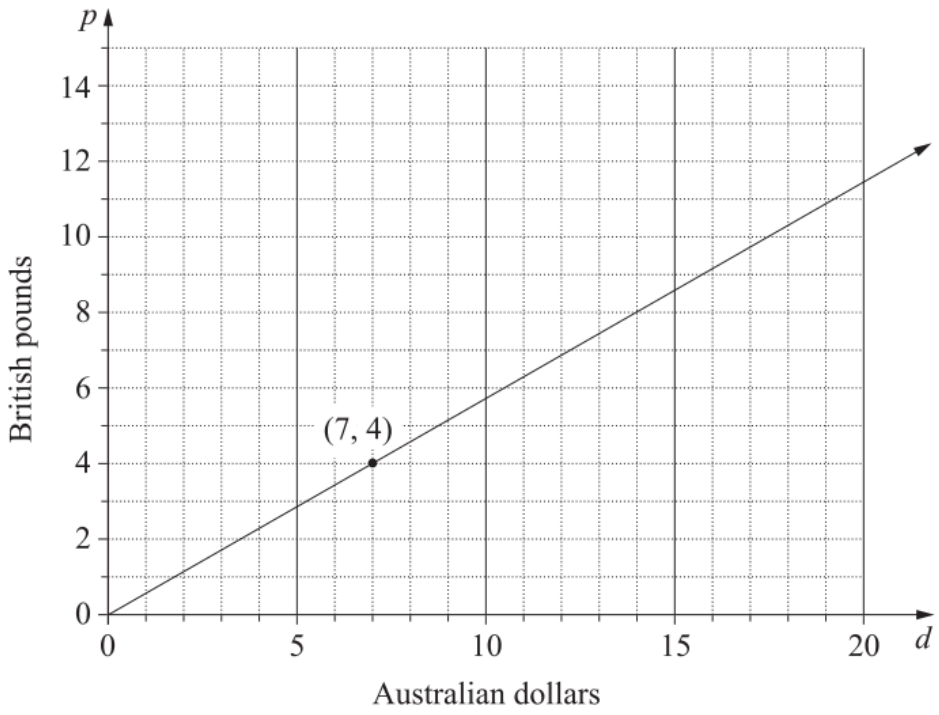
2019 HSC Mathematics Standard 2 · Question 34

Algebra A2 - Linear relationships · 3 marks

42%

state mean mark, part (a)

The relationship between British pounds (p) and Australian dollars (d) on a particular day is shown in the graph.



a. Write the direct variation equation relating British pounds to Australian dollars in the form $p = md$. Leave m as a fraction. *(1 mark)*

.....

.....

b. The relationship between Japanese yen (y) and Australian dollars (d) on the same day is given by the equation $y = 76d$.

Convert 93 100 Japanese yen to British pounds. *(2 marks)*

WORKING

#34

2011 HSC General Mathematics · Question 27d

Financial F5 - Annuities · 4 marks

42%
state mean mark

Josephine invests $50 000 for 15 years, at an interest rate of 6% per annum, compounded annually.

Emma invests $500 at the end of each month for 15 years, at an interest rate of 6% per annum, compounded monthly.

Financial gain is defined as the difference between the final value of an investment and the total contributions.

Who will have the better financial gain after 15 years? Using the Table below* and appropriate formulas, justify your answer with suitable calculations. ***(4 marks)***

Future value of an annuity of $1				
Period	0.50%	1%	2%	3%
120	163.8793	230.0387	488.2582	1123.6996
180	290.8187	499.5802	1716.0416	6783.4453
240	462.0409	989.2554	5744.4368	40128.4209

**Question adjusted to reflect the updated syllabus.*

WORKING

#35

2011 HSC General Mathematics · Question 27c

Statistics S5 - The normal distribution · 3 marks

42%

state mean mark, part (ii)

Two brands of light bulbs are being compared. For each brand, the life of the light bulbs is normally distributed.

Life of light bulbs (in hours)

	Mean	*Standard deviation*
Brand A	450	25
Brand B	500	50

i. One of the Brand B light bulbs has a life of 400 hours.
What is the z-score of the life of this light bulb? ***(1 mark)***

ii. A light bulb is considered defective if it lasts less than 400 hours. The following claim is made:
'Brand A light bulbs are more likely to be defective than Brand B light bulbs.'
Is this claim correct? Justify your answer, with reference to z-scores or standard deviations or the normal distribution. ***(2 marks)***

WORKING

#36

2010 HSC General Mathematics · Question 24c

Statistics S5 - The normal distribution · 4 marks

42%

state mean mark, part (ii)

The marks in a class test are normally distributed. The mean is 100 and the standard deviation is 10.

(i) Jason's mark is 115. What is his z-score? ***(1 mark)***

(ii) Mary has a z-score of 0. What mark did she achieve in the test? ***(1 mark)***

(iii) What percentage of marks lie between 80 and 110?

You may assume the following:

- 68% of marks have a z-score between -1 and 1
- 90% of marks have a z-score between -2 and 2
- 99.7% of marks have a z-score between -3 and 3. ***(2 marks)***

WORKING

#38 2014 HSC Mathematics General 2 · Question 21

Financial F5 - Annuities · Multiple choice · 1 mark

43%
state mean mark

A table of future value interest factors is shown.

Table of future value interest factors

Period	*Interest rate per period*				
	1%	*2%*	*3%*	*4%*	*5%*
1	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0100	2.0200	2.0300	2.0400	2.0500
3	3.0301	3.0604	3.0909	3.1216	3.1525
4	4.0604	4.1216	4.1836	4.2465	4.3101

A certain annuity involves making equal contributions of $25 000 into an account every 6 months for 2 years at an interest rate of 4% per annum.

Based on the information provided, what is the future value of this annuity?

(A) $50 500

(B) $51 000

(C) $103 040

(D) $106 162

Answer: A B C D

WORKING

#39 2012 HSC General Mathematics · Question 11

Statistics S4 - Bivariate data analysis · Multiple choice · 1 mark

43%
state mean mark

Which of the following relationships would most likely show a negative correlation?

A. The population of a town and the number of hospitals in that town.

B. The hours spent training for a race and the time taken to complete the race.

C. The price per litre of petrol and the number of people riding bicycles to work.

D. The number of pets per household and the number of computers per household.

Answer: A B C D

WORKING

#40

2019 HSC Mathematics Standard 2 · Question 38

Statistics S5 - The normal distribution · 2 marks

44%
state mean mark

In a particular country, the birth weight of babies is normally distributed with a mean of 3000 grams. Is is known that 95% of these babies have a birth weight between 1600 grams and 4400 grams.

One of these babies has a birth weight of 3497 grams. What is the z-score of this baby's birth weight? ***(2 marks)***

WORKING

#41

2015 HSC Mathematics General 2 · Question 28b

Statistics S5 - The normal distribution · 2 marks

44%
state mean mark

The results of two tests are normally distributed. The mean and standard deviation for each test are displayed in the table.

	Mathematics	English
$\bar{x}$	70	75
s	6.5	8

Kristoff scored 74 in Mathematics and 80 in English. He claims that he has performed better in English.

Is Kristoff correct? Justify your answer using appropriate calculations. ***(2 marks)***

WORKING

#42

2013 HSC General Mathematics · Question 26e

Financial F4 - Investments and loans · 2 marks

44%
state mean mark

Kimberley has invested $3500.

Interest is compounded half-yearly at a rate of 2% per half-year.

Compounded values of $1

Period	*Interest rate per period*					
	1%	2%	3%	4%	5%	6%
1	1.010	1.02	1.03	1.04	1.05	1.06
2	1.020	1.040	1.061	1.082	1.103	1.124
3	1.030	1.061	1.093	1.125	1.158	1.191
4	1.041	1.082	1.126	1.170	1.216	1.262
5	1.051	1.104	1.159	1.217	1.276	1.338
6	1.062	1.126	1.194	1.265	1.340	1.419
7	1.072	1.149	1.230	1.316	1.407	1.504
8	1.083	1.172	1.267	1.369	1.477	1.594

Use the table to calculate the value of her investment at the end of 4 years. ***(2 marks)***

WORKING

#43

2012 HSC General Mathematics · Question 29b

Statistics S5 - The normal distribution · 2 marks

44%
state mean mark

A machine produces nails. When the machine is set correctly, the lengths of the nails are normally distributed with a mean of 6.000 cm and a standard deviation of 0.040 cm.

To confirm the setting of the machine, three nails are randomly selected. In one sample the lengths are 5.950, 5.983 and 6.140.

The setting of the machine needs to be checked when the lengths of two or more nails in a sample lie more than 1 standard deviation from the mean.

Does the setting on the machine need to be checked? Justify your answer with suitable calculations.

(2 marks)

WORKING

#44 2014 HSC Mathematics General 2 · Question 24

Statistics S5 - The normal distribution · Multiple choice · 1 mark

46%
state mean mark

The weights of 10 000 newborn babies in NSW are normally distributed. These weights have a mean of 3.1 kg and a standard deviation of 0.35 kg.

How many of these newborn babies have a weight between 2.75 kg and 4.15 kg?

(A) 4985

(B) 6570

(C) 8370

(D) 8385

Answer: A B C D

WORKING

#45 2018 HSC Mathematics General 2 · Question 23

Statistics S5 - The normal distribution · Multiple choice · 1 mark

47%
state mean mark

A set of data is normally distributed with a mean of 48 and a standard deviation of 3.

Approximately what percentage of the scores lies between 39 and 45?

A. 15.85%

B. 31.7%

C. 47.5%

D. 49.85%

Answer: A B C D

WORKING

#46

2017 HSC Mathematics General 2 · Question 27a

Statistics S1 - Data analysis · 2 marks

47%

state mean mark, part (ii)

Jamal surveyed eight households in his street. He asked them how many kilolitres (kL) of water they used in the last year. Here are the results.

$$220, 105, 101, 450, 37, 338, 151, 205$$

i. Calculate the mean of this set of data. *(1 mark)*

.....

ii. What is the standard deviation of this set of data, correct to one decimal place? *(1 mark)*

WORKING

#47

2017 HSC Mathematics General 2 · Question 29c

Probability (now Statistics S2) · 4 marks

47%

state mean mark, part (ii)

A group of Year 12 students was surveyed. The students were asked whether they live in the city or the country. They were also asked if they have ever waterskied.

The results are recorded in the table.

	Have waterskied	Have never waterskied
Live in the city	150	2500
Live in the country	70	800

(i) A person is selected at random from the group surveyed. Calculate the probability that the person lives in the city and has never waterskied. ***(2 marks)***

(ii) A newspaper article claimed that Year 12 students who live in the country are more likely to have waterskied than those who live in the city.
Is this true, based on the survey results? Justify your answer with relevant calculations. ***(2 marks)***

WORKING

#48

2014 HSC Mathematics General 2 · Question 30a

Financial F4 - Investments and loans · 3 marks

49%
state mean mark

Chandra and Sascha plan to have $20 000 in an investment account in 15 years time for their grandchild's university fees.

The interest rate for the investment account will be fixed at 3% per annum compounded monthly.

Calculate the amount that they will need to deposit into the account now in order to achieve their plan. ***(3 marks)***

WORKING

#49 2015 HSC Mathematics General 2 · Question 20

Statistics S5 - The normal distribution · Multiple choice · 1 mark

50%
state mean mark

A machine produces cylindrical pipes. The mean of the diameters of the pipes is 8 cm and the standard deviation is 0.04 cm.

Assuming a normal distribution, what percentage of cylindrical pipes produced will have a diameter less than 7.96 cm?

A. 16%

B. 32%

C. 34%

D. 68%

Answer: (A) (B) (C) (D)

WORKING

#50 2011 HSC General Mathematics · Question 20

Algebra A4 - Types of relationships · Multiple choice · 1 mark

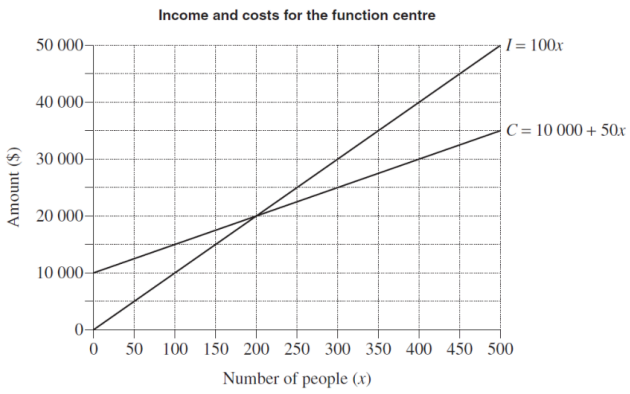
50%
state mean mark

A function centre hosts events for up to 500 people. The cost C, in dollars, for the centre to host an event, where x people attend, is given by:

$$C = 10\ 000 + 50x$$

The centre charges \$100 per person. Its income I, in dollars, is given by:

$$I = 100x$$



How much greater is the income of the function centre when 500 people attend an event, than its income at the breakeven point?

A. \$15 000

B. \$20 000

C. \$30 000

D. \$40 000

Answer: (A) (B) (C) (D)

WORKING

Worked solutions and markers' comments (rank order)

#1 2013 HSC General Mathematics · Q27c · state mean 1%

i. Since cumulative frequency $= 52$ at max

$\Rightarrow$ Lower quartile at week $= \dfrac{52}{4} = 13$

Lower quartile $= 190$ (from graph)

$\Rightarrow$ Upper quartile at week $= 3 \times 13 = 39$

Upper quartile $= 470$

$\therefore$ IQR $= 470 - 190$

$= 280$

♦♦♦ *Mean mark 13%*
COMMENT: Finding median and quartile values from cumulative frequency charts is an important skill that is often examined.

ii. Oscar is not correct because the data is grouped.

He could only say the retailer sold between 250 and 350 units in 6 of the weeks in 2012.

♦♦♦ *This question proved a beast, with a mean mark of 1%. Toughest question in the 2013 exam!*

#2 2011 HSC General Mathematics · Q25a · state mean 7%

(i) Categorical

(ii) How many outgoing calls do you make per day?
(Ensure it can be answered with a numerical score.)

(iii) The method could be to work out how many students are in each year and ask 10% of the students in each year. (Note the sample of students in each year must be proportional to their percentage in the population).

♦♦♦ *Mean mark 7%. Toughest mark to get in the 2011 exam!*
COMMENT: Know and be able to describe random, systematic and stratified sampling!

(iv) A census would involve all high school students in NSW.

♦♦♦ *Mean mark 18%.*
MARKER'S COMMENT: A ***specific*** population needed (i.e. high school students).

#3 2017 HSC Mathematics General 2 · Q29d · state mean 13%

i. Median $= \dfrac{15\text{th} + 16\text{th score}}{2}$

$= \dfrac{4+8}{2}$

$= 6$

♦ *Mean mark 50%.*

ii. Lower limit $= 5.4 - 4.22 = 1.18$

Upper limit $= 5.4 + 4.22 = 9.62$

$\therefore$ Percentage in between

$= \dfrac{13}{30} \times 100$

$= 43.33\ldots$

$= 43\%$ (nearest %)

♦♦ *Mean mark 34%.*

iii. The statement assumes the data is normally distributed which is not the case here.

♦♦♦ *Mean mark 13%.*

#4 2019 HSC Mathematics Standard 2 · Q15 · state mean 17%

68% of marks lie between 64 and 76 (mean $\pm$ 1 σ).

50% of marks lie between Q_1 and Q_3.

$\Rightarrow A$

♦♦♦ *Mean mark 17%.*

#5 2019 HSC Mathematics Standard 2 · Q10 · state mean 18%

Train or bus delay (%)

$= 92 - 86$

$= 6\%$

$\Rightarrow A$

♦♦♦ *Mean mark 18%.*

#6 2009 HSC General Mathematics · Q28b · state mean 18%

i. The correlation between height and mass is positive and strong.

♦ *Mean mark 48%.*

ii. Using $P_1(40, 1.2)$ and $P_2(80, 10.4)$

$$\text{Gradient} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{10.4 - 1.2}{80 - 40} = \frac{9.2}{40} = 0.23$$

Line passes through $P_1(40, 1.2)$

Using $y - y_1 = m(x - x_1)$

$y - 1.2 = 0.23(x - 40)$

$y - 1.2 = 0.23x - 9.2$

$y = 0.23x - 8$

$\therefore$ Equation of the line is $M = 0.23H - 8$

♦♦♦ *Mean mark 18%.*
MARKER'S COMMENT: Many students had difficulty due to the fact the horizontal axis started at $H = 40\text{cm}$ and not the origin.

#7 2013 HSC General Mathematics · Q29b · state mean 21%

(i) $\sigma = 5.2201\ldots$

$= 5.2$ (to 1 d.p.)

♦ *Mean mark 39%*
COMMENT: Make sure you are confident with this function on your calculator!

(ii) $z = \dfrac{x - \mu}{\sigma}$

z-score (1st test) $= \dfrac{62 - 68.5}{5.2}$

$= -1.25$

Find the mark in 2nd test with a z-score of -1.25

$-1.25 = \dfrac{x - 74.4}{12.4}$

$x - 74.4 = -15.5$

$x = 58.9$

$\therefore$ Ali needs to score 58.9

♦♦ *Mean mark 21%.*
MARKER'S COMMENT: When "performance relative to the class is maintained", ***z*-scores** are the same in each test.

#8 2017 HSC Mathematics General 2 · Q27c · state mean 22%

(i) FV factor $= 5.4163$

$\therefore$ FV of Annuity $= 12\,000 \times 5.4163$

$= \$64\,995.60$

(ii) Interest earned $=$ FV $-$ total repayments

$= 64\,995.60 - (5 \times 12\,000)$

$= \$4995.60$

♦♦ *Mean mark part (ii) 22%.*
COMMENT: A very poorly answered question dealing with a core concept in this area.

#9 2017 HSC Mathematics General 2 · Q30d · state mean 23%

(i) Speed$(y) = 2.125(5) + 2.0375$

$= 12.6625$ km/h

(ii) Using the formula sheet:

$$\text{gradient} = r \times \frac{\text{std dev}(y)}{\text{std dev}(x)}$$

$2.125 = r \times \dfrac{2}{0.8}$

$\therefore r = \dfrac{2.125 \times 0.8}{2}$

$= 0.85$

♦♦ *Mean mark 23%.*

#10 2013 HSC General Mathematics · Q20 · state mean 24%

Since 68% between $z = 1$ and -1

$\Rightarrow$ 34% between $z = 0$ and 1

Likewise, 95% between $z = 2$ and -2

$\Rightarrow$ 47.5% between $z = 0$ and 2

$\therefore$ % between $z = 1$ and 2

$= 47.5 - 34$

$= 13.5\%$

$\therefore$ Students between $z = 1$ and $z = 2$

$= 13.5\% \times 60\ 000$

$= 8100$

$\Rightarrow A$

♦♦ *Mean mark 24%*

#11 2012 HSC General Mathematics · Q9 · state mean 24%

4% annual

$\Rightarrow \dfrac{4\%}{4} = 1\%$ compounded each quarter

$\Rightarrow n = 8$ (8 quarters in 2 years)

$\therefore$ Factor $= 1.083$ (from table)

$\Rightarrow C$

♦♦♦ *Mean mark 24%. The lowest MC mean mark in the 2012 exam.*
COMMENT: A process to follow: 1- convert the annual rate to the rate per compounding period. 2- calculate the number of compounding periods.

#12 2016 HSC Mathematics General 2 · Q7 · state mean 26%

Categorical and nominal data is qualitative and not ordered.

$\Rightarrow A$

♦♦ *Mean mark 26%.*

#13 2019 HSC Mathematics Standard 2 · Q42 · state mean 27%

Annuity compounding factor (4% for 6 years) $= 6.633$

$\therefore$ Value after 6 years $= 2000 \times 6.633$

$= \$13\ 266.00$

At the end of 7th year:

Value $= 13\ 266 \times 1.05 + 2000$

$= 13\ 929.30 + 2000$

$= \$15\ 929.30$

♦♦ *Mean mark 27%.*

#14 2018 HSC Mathematics General 2 · Q26e · state mean 27%

42 data points $\Rightarrow$ median $= \dfrac{21\text{st} + 22\text{nd}}{2}$

$Q_1 = 11\text{th data point} = 3$

$Q_3 = 32\text{nd data point} = 5$

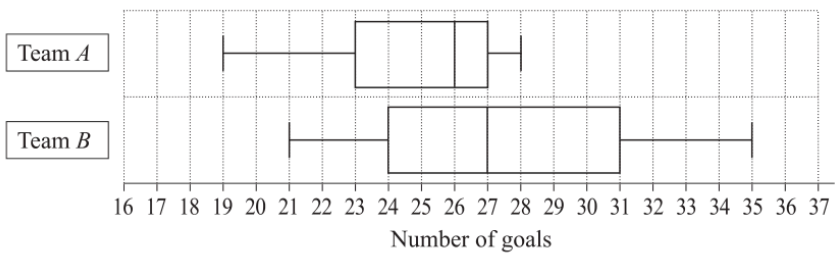
$\therefore$ IQR $= 5 - 3$

$= 2$

♦♦ *Mean mark 27%.*

#15 2019 HSC Mathematics Standard 2 · Q39 · state mean 28%

Team A: High $= 28$, Low $= 19$, $Q_1 = 23$, $Q_3 = 27$, Median $= 26$



Team A's distribution is negatively skewed while Team B's distribution is slightly positively skewed.

The standard deviation of Team A's distribution is smaller than Team B, as both its IQR and range is smaller.

Team B is a more successful team at scoring goals as each value in its 5-point summary is higher than Team A's equivalent value.

♦♦ *Mean mark 28%.*

#16 2011 HSC General Mathematics · Q23c · state mean 28%

Interest rate $= 10\%$ pa $= 5\%$ per 6 months

Period $= 6$ (6 x 6 months in 3 years)

$\Rightarrow$ Table value $= 1.340$

$\therefore$ Value of investment $= 5000 \times 1.34$

$= \$6700$

♦♦ *Mean mark 28%*
MARKER'S COMMENT: Remember that the number of periods is the number of "compounding periods" and when asked to use the table, *use the table!*

#17 2009 HSC General Mathematics · Q21 · state mean 28%

If $\bar{x}$ of 10 scores $= 14$

$\Rightarrow$ Sum of 10 scores $= 10 \times 14 = 140$

With 2 additional scores, $\bar{x} = 16$

$\Rightarrow$ Sum of 12 scores $= 12 \times 16 = 192$

$\therefore$ Value of 2 extra scores $= 192 - 140$

$= 52$

$\therefore$ Mean of 2 extra scores $= \dfrac{52}{2} = 26$

$\Rightarrow D$

♦♦♦ *Mean mark 28%.*

#18 2009 HSC General Mathematics · Q27a · state mean 31%

(i) Table factor when $n = 6, \quad r = 3\%$

$\Rightarrow 6.4684$

$\therefore FV = 5000 \times 6.4684$

$= \$32\,342$

(ii) Table factor when $n = 7, \quad r = 5\%$

$\Rightarrow 8.1420$

Let $A =$ annuity

$FV = A \times 8.1420$

$A = \dfrac{FV}{8.1420}$

$= \dfrac{407\,100}{8.1420}$

$= \$50\,000$

♦ *Mean mark 45%*
MARKER'S COMMENT: A common error was to multiply \$407 100 by 8.1420 rather than divide.

(iii) $n = 8$ (8 quarters in 2 years)

$r = \dfrac{4\%}{4} = 1\%$ per quarter

$\therefore$ Table factor $\Rightarrow 8.2857$

$FV = 1000 \times 8.2857$

$= 8285.70$

Interest $= FV(\text{annuity}) - \text{Principal}$

$= 8285.70 - (8 \times 1000)$

$= 285.70$

$\therefore$ Interest earned is \$285.70

♦♦ *Mean mark 31%*
MARKER'S COMMENT: When questions asked for the interest paid on annuities, remember to subtract the total principal amounts contributed.

#19 2020 HSC Mathematics Standard 2 · Q7 · state mean 32%

Positive skew occurs when the tail on the histogram is longer on the right-hand (positive) side.

$\Rightarrow A$

♦♦ *Mean mark 32%.*

#20 2016 HSC Mathematics General 2 · Q30d · state mean 32%

i. $\mu = 88 - 2.4\sigma$

ii. $\mu = 52 + 1.2\sigma \ldots (1)$

$\mu = 88 - 2.4\sigma \ldots (2)$

Subtract $(2) - (1)$:

$0 = 36 - 3.6\sigma$

$3.6\sigma = 36$

$\therefore \sigma = 10$

Substitute $\sigma = 10$ into (1):

$\therefore \mu = 52 + 1.2 \times 10$

$= 64$

♦♦ *Mean mark part (ii) 32%.*
COMMENT: Important to recognise this as a simultaneous equation problem and **not** a normal distribution one.

#21 2018 HSC Mathematics General 2 · Q19 · state mean 33%

4% annual

$\Rightarrow \dfrac{4\%}{4} = 1\%$ compounded quarterly

$\Rightarrow n = 8$

$\Rightarrow$ Factor $= 1.0829$

$\therefore$ Minimum sum $= 21\,000 \div 1.0829$

$\Rightarrow$ D

♦♦ *Mean mark 33%.*

#22 2016 HSC Mathematics General 2 · Q29e · state mean 33%

(i) 68 years

(ii) Using (1900,60), (1980,80)

$\text{Gradient} = \dfrac{y_2 - y_1}{x_2 - x_1}$

$= \dfrac{80 - 60}{1980 - 1900}$

$= 0.25$

$\therefore$ After 1900, life expectancy increases by 0.25 years for each year later that someone is born.

♦♦ *Mean mark part (ii) 33%*

#23 2015 HSC Mathematics General 2 · Q30c · state mean 33%

i. $N = 74, r = 0.0080$

$PV(\text{annuity})$ table factor $= 55.68446$

$\therefore PV$ of annuity

$= \$200 \times 55.68446$

$= \$11\,136.892$

$= \$11\,136.89$ (nearest cent)

♦ *Mean mark 48%.*

ii. Over 6 years

$N = 6 \times 12 = 72$ months

$r = \dfrac{10.8}{12} = 0.9\% = 0.009$

$PV(\text{annuity})$ table factor $= 52.82118$

Let \$M = monthly repayment

Loan $= PV$ of annuity

$\therefore \$21\,500 = \text{M} \times 52.82118$

$\text{M} = \$407.033\ldots$

$= \$407$ (nearest dollar)

♦♦ *Mean mark 33%.*

#24 2016 HSC Mathematics General 2 · Q23 · state mean 34%

$P(\text{Smoker or a male})$

$= \dfrac{\text{Total males + female smokers}}{\text{Total surveyed}}$

$= \dfrac{264 + 68}{485}$

$= 0.684\ldots$

$\Rightarrow C$

♦♦ Mean mark 34%.

#25 2011 HSC General Mathematics · Q25d · state mean 34%

i. Outlier is 71

ii. Lower quartile $= 9$ (4th female data point)

Upper quartile $= 20$ (11th female data point)

$\therefore$ Interquartile range (female) $= 20 - 11 = 9$

♦♦ *Mean mark 34%*
COMMENT: Ensure you can quickly and accurately find quartile values using stem and leaf graphs!

#26 2013 HSC General Mathematics · Q28b · state mean 38%

(i) $\text{Gradient} = \dfrac{\text{rise}}{\text{run}}$

$$= \frac{176-146}{16-11}$$

$$= \frac{30}{5}$$

$$= 6$$

(ii) Males should grow 6cm per year between the ages 11-16.

♦♦ *Mean marks of 38%, 26% and 25% respectively for parts (i)-(iii).*
MARKER'S COMMENT: Interpreting gradients has been consistently examined in recent history and almost always poorly answered.

(iii) Gradient $= 6$

Passes through $(11, 146)$

$$y - y_1 = m(x - x_1)$$

$$h - 146 = 6(a - 11)$$

$$= 6a - 66$$

$$\therefore h = 6a + 80$$

(iv) If $a = 17$

$$h = (6 \times 17) + 80$$

$$= 182$$

$\therefore$ A typical 17 year old is expected to be 182cm

(v) People slow and eventually stop growing after they become adults.

#27 2015 HSC Mathematics General 2 · Q19 · state mean 39%

In 1975, her life expectancy

$$= \text{age} + \text{remaining years}$$

$$= 45 + 34$$

$$= 79$$

♦ *Mean mark 39%.*

In 1995, her life expectancy

$$= 65 + 19.8$$

$$= 84.8$$

$$\therefore \text{Difference} = 84.8 - 79$$

$$= 5.8 \text{ years}$$

$\Rightarrow D$

#28 2019 HSC Mathematics Standard 2 · Q23 · state mean 40%

a. Use "A + Bx" function (fx-82 calc):

$$r = 0.9811\ldots$$

$$= 0.98 \quad (2 \text{ d.p.})$$

♦ *Mean mark 40%.*
COMMENT: Issues here? YouTube has short and excellent help videos – search your calculator model and topic – eg. "fx-82 correlation".

b. Direction: positive

Strength: strong

c. $\text{Height} = 0.866 \times 143 + 23.7$

$$= 147.538 \text{ cm}$$

#29 2018 HSC Mathematics General 2 · Q26a · state mean 40%

$$\text{Rel Freq} = \frac{\text{number of 3's rolled}}{\text{total rolls}}$$

$$= \frac{48}{150}$$

$$= \frac{8}{25}$$

♦ *Mean mark 40%.*

#30 2016 HSC Mathematics General 2 · Q28d · state mean 40%

$\text{Periods} = 6 \times 4 = 24$

$\text{Interest rate} = \dfrac{1}{4} \times 3 = 0.75\%$

♦ *Mean mark 40%.*

$\Rightarrow$ Table factor $= 0.0382$

(i.e. 3.82 cents contributed per quarter $= \$1$ after 6 years)

$\therefore$ Quarterly contribution

$$= 75\,000 \times 0.0382$$

$$= \$2865$$

#31 2013 HSC General Mathematics · Q23 · state mean 41%

Interest: 3% p.a $\Rightarrow$ 1.5% per 6 months

After 2 years,

♦ *Mean mark 41%.*

$\text{Value of 1st deposit} = 1200(1.015)^3 = 1254.81$

$\text{Value of 2nd deposit} = 1200(1.015)^2 = 1236.27$

$\text{Value of 3rd deposit} = 1200(1.015) = 1218$

$\text{Value of 4th deposit} = 1200$

$\therefore$ Amount in account after 2 years

$$= 1254.81 + 1236.27 + 1218 + 1200$$

$$= \$4909.08$$

$\Rightarrow A$

#32 2009 HSC General Mathematics · Q25d · state mean 41%

(i) $\mu = 25.8 \quad \sigma = 4.2$

Using $z = \dfrac{x - \mu}{\sigma}$

$$-1 = \frac{x - 25.8}{4.2}$$

$$x - 25.8 = -4.2$$

$$x = 21.6^\circ$$

$\therefore 21.6^\circ$ has a z-score of -1

(ii) z-score of $21.6 = -1$

Find z-score of 38.4

$$z\,(38.4) = \frac{38.4 - 25.8}{4.2} = 3$$

♦ Mean mark 41%
MARKER'S COMMENT: Many students failed to use the answer to (d)(i), costing them valuable exam time.

68% of scores are between $z = -1$ and 1

$\Rightarrow$ 34% are between $z = -1$ and 0

99.7% of scores are between $z = -3$ and 3

$\Rightarrow$ 49.85% are between $z = 0$ and 3

$\therefore$ % Temps between 21.6° and 38.4°

$$= 34\% + 49.85\%$$

$$= 83.85\%$$

#33 2019 HSC Mathematics Standard 2 · Q34 · state mean 42%

a. $m = \dfrac{\text{rise}}{\text{run}} = \dfrac{4}{7}$

$p = \dfrac{4}{7}d$

♦ *Mean mark 42%.*

b. Yen to Australian dollars:

$$y = 76d$$
$$93\ 100 = 76d$$
$$d = \frac{93\ 100}{76}$$
$$= 1225$$

Aust dollars to pounds:

$$p = \frac{4}{7} \times 1225$$
$$= 700 \text{ pounds}$$

$\therefore 93\ 100 \text{ Yen} = 700 \text{ pounds}$

#34 2011 HSC General Mathematics · Q27d · state mean 42%

Josephine

$$\text{Investment} = 50\,000(1 + 0.06)^{15}$$
$$= 50\,000(1.06)^{15}$$
$$= \$119\,827.91$$

$$\text{Financial gain} = 119\,827.91 - 50\,000$$
$$= \$69\,827.91$$

Emma

Monthly interest rate $= 6\% \div 12 = 0.5\%$

\# Monthly Payments $= 12 \times 15 = 180$

$\Rightarrow$ Annuity Factor $= 290.8187$ (from Table)

$$\text{Investment} = 500 \times 290.8187$$
$$= \$145\,409.35$$

$$\text{Financial gain} = 145\,409.35 - \text{total contributions}$$
$$= 145\,409.35 - (500 \times 12 \times 15)$$
$$= 145\,409.35 - 90\,000$$
$$= \$55\,409.35$$

$\therefore$ Josephine will have the better financial gain.

♦ *Mean mark 42%*
COMMENT: Note that compound interest vs annuity comparisons are commonly tested.

#35 2011 HSC General Mathematics · Q27c · state mean 42%

i. z-score of Brand B bulb (400 hrs)

$$= \frac{x - \mu}{\sigma}$$
$$= \frac{400 - 500}{50}$$
$$= -2$$

ii. z-score of Brand A bulb (400 hours)

$$= \frac{400 - 450}{25}$$
$$= -2$$

Since the z-score for both brands is -2,
they are equally likely to be defective.

$\therefore$ The claim is incorrect.

♦ *Mean mark 42%*
MARKER'S COMMENT: A number of students found drawing normal curves in their solution advantageous.

#36 2010 HSC General Mathematics · Q24c · state mean 42%

(i) Given $\mu = 100,\ \sigma = 10$

If mark is 115,

$$z\text{-score} = \frac{115 - \mu}{\sigma}$$
$$= \frac{115 - 100}{100}$$
$$= 1.5$$

MARKER'S COMMENT: Too may students had calculator errors in this question, giving away easy marks. *BE CAREFUL!*

(ii) z-score $= 0$ when mark equals the mean

$\therefore$ Mary's score was 100

(iii) z-score of $110 = \dfrac{110 - 100}{10} = 1$

z-score of $80 = \dfrac{80 - 100}{10} = -2$

68% of marks lie between $z = -1$ and 1
$\Rightarrow$ 34% lie between $z = 0$ and 1
95% of marks lie between $z = -2$ and 2
$\Rightarrow$ 47.5% lie between $z = -2$ and 0

$\therefore$ % marks between 80 and 110
$= 34\% + 47.5\%$
$= 81.5\%$

♦ *Mean mark 42%*

#37 2016 HSC Mathematics General 2 · Q21 · state mean 43%

Using the centre of each class interval:

$$\text{Mean} = \frac{3 \times 3 + 8 \times 6 + 13 \times 8 + 18 \times 9}{3 + 6 + 8 + 9}$$
$$= 12.42\ldots$$

$\Rightarrow D$

♦ *Mean mark 43%.*

#38 2014 HSC Mathematics General 2 · Q21 · state mean 43%

4 contributions of \$25 000 made.

Annuity period $= 6$ months

$$\text{Rate (per annuity period)} = \frac{4\%}{2} = 2\%$$

\# Periods $= 4$ (4 x 6 months $= 2$ years)

Table value $= 4.1216$

$\therefore$ Annuity Value $= 4.1216 \times 25\,000 = \$103\,040$

$\Rightarrow C$

♦ *Mean mark 43%*

#39 2012 HSC General Mathematics · Q11 · state mean 43%

Increased hours training should reduce the time to complete a race.

$\Rightarrow B$

♦ *Mean mark 43%.*

#40 2019 HSC Mathematics Standard 2 · Q38 · state mean 44%

95% babies within $1600 - 4400$

i.e. $3000 \pm 2\sigma = 1600 - 4400$

$$2\sigma = 4400 - 3000$$
$$= 1400$$
$$\sigma = 700$$

$$\therefore z\text{-score}\ (3497) = \frac{x - \mu}{\sigma}$$
$$= \frac{3497 - 3000}{700}$$
$$= 0.71$$

♦ *Mean mark 44%.*

#41 2015 HSC Mathematics General 2 · Q28b · state mean 44%

In Maths

$$z\text{-score}(74) = \frac{x-\mu}{\sigma}$$
$$= \frac{74-70}{6.5}$$
$$= 0.6153\ldots$$

♦ *Mean mark 44%.*

In English

$$z\text{-score}(80) = \frac{80-75}{8}$$
$$= 0.625$$

$\Rightarrow$ Kristoff's z-score in English is higher than his z-score in Maths.

$\therefore$ He is correct. He performed better in English.

#42 2013 HSC General Mathematics · Q26e · state mean 44%

$r = 2\%$ per half-year

$n = 8$ (8 half-years in 4 years)

$\Rightarrow$ Table Factor $= 1.172$

Investment $= 3500 \times 1.172$

$= \$4102$

$\therefore$ After 4 years, investment value is \$4102

♦ *Mean mark 44%*
COMMENT: Structure your answer: 1-Find the interest rate per compounding period (same in this case). 2-Find the number of *compounding periods*.

#43 2012 HSC General Mathematics · Q29b · state mean 44%

$\mu = 6.000, \quad \sigma = 0.040$

Limits for $\pm 1\,\sigma$:

$6.000 + 0.040 = 6.040$ (upper)

$6.000 - 0.040 = 5.960$ (lower)

Chosen nails: $5.950 \Rightarrow$ outside limits

$5.983 \Rightarrow$ inside

$6.140 \Rightarrow$ outside

Since 2 nails are outside 1 σ limits

$\therefore$ Settings need to be checked.

♦ *Mean mark 44%*

#44 2014 HSC Mathematics General 2 · Q24 · state mean 46%

Find z-scores of 2.75 and 4.15 kg

$$z\,(2.75) = \frac{x-\mu}{5} = \frac{2.75-3.1}{0.35} = -1$$

$$z\,(4.15) = \frac{4.15-3.1}{0.35} = 3$$

68% between $z = -1$ and 1

$\Rightarrow$ 34% between $z = -1$ and 0

99.7% between $z = -3$ and 3

$\Rightarrow$ 49.85% between $z = 0$ and 3

$\therefore$ % with z-scores between -1 and 3

$= 34 + 49.85$

$= 83.85\%$

$\therefore$ # Babies between 2.75 kg and 4.15 kg

$= 83.85\% \times 10\,000$

$= 8385$

$\Rightarrow D$

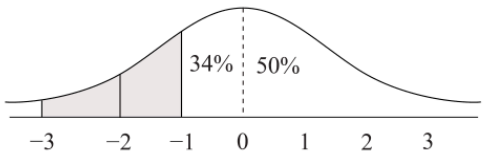
♦ *Mean mark 46%*

#45 2018 HSC Mathematics General 2 · Q23 · state mean 47%

Given $\mu = 48, \ \sigma = 3$

$$z\text{-score}\ (39) = \frac{x-\mu}{\sigma}$$
$$= \frac{39-48}{3}$$
$$= -3$$

$$z\text{-score}\ (45) = \frac{45-48}{3}$$
$$= -1$$



$\therefore$ Scores between 39 and 45

$\approx 16\%$

$\Rightarrow A$

Note that % of scores below 39 is very small.

♦ *Mean mark 47%.*

#46 2017 HSC Mathematics General 2 · Q27a · state mean 47%

i. Mean $= (220 + 105 + 101 + 450 + 37 + 338 + 151 + 205) \div 8$

$= 200.875$

ii. Std Dev $= 127.357\ldots$ (by calc)

$= 127.4$ (1 d.p.)

♦ *Mean mark part (ii) 47%.*
IMPORTANT: The *population* standard deviation is required here.

#47 2017 HSC Mathematics General 2 · Q29c · state mean 47%

(i) $P = \dfrac{\text{live in city, not skied}}{\text{total surveyed}}$

$= \dfrac{2500}{3520}$

$= \dfrac{125}{176}$

(ii) $P(\text{live in country, skied}) = \dfrac{70}{(70+800)}$

$= 0.0804\ldots$

$= 8\%$

$P(\text{live in city, skied}) = \dfrac{150}{(150+2500)}$

$= 0.0566$

$= 6\%$

Since $8\% > 6\%$, the statement is true.

♦ *Mean mark part (ii) 47%.*

#48 2014 HSC Mathematics General 2 · Q30a · state mean 49%

$FV = \$20\,000, \quad n = 15 \times 12 = 180,$

$r = \dfrac{0.03}{12} = 0.0025$

$FV = PV(1+r)^n$

$20\,000 = PV(1+0.0025)^{180}$

$PV = \dfrac{20\,000}{(1.0025)^{180}}$

$= 12\,759.73\ldots$

$\therefore$ They need to deposit \$12 760 (nearest \$)

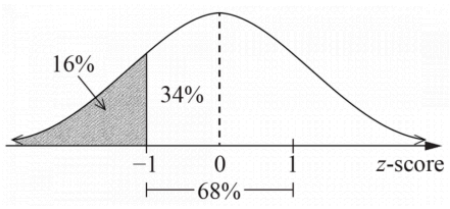
♦ *Mean mark 49%*

#49 2015 HSC Mathematics General 2 · Q20 · state mean 50%

$\mu = 8 \text{ cm} \quad s = 0.04 \text{ cm}$

$$
\begin{aligned}
z\text{-score}(7.96) &= \frac{x - \mu}{\sigma} \\
&= \frac{7.96 - 8}{0.04} \\
&= -1
\end{aligned}
$$

♦ *Mean mark 50%.*



$\therefore$ % of pipes with a diameter less than 7.96 cm.

$= 50\% - 34\%$

$= 16\%$

$\Rightarrow A$

#50 2011 HSC General Mathematics · Q20 · state mean 50%

When $x = 500,\ I = 100 \times 500 = \$50\ 000$

Breakeven when $x = 200$ (from graph)

When $x = 200,\ I = 100 \times 200 = \$20\ 000$

$$
\begin{aligned}
\text{Difference} &= 50\ 000 - 20\ 000 \\
&= \$30\ 000
\end{aligned}
$$

$\Rightarrow C$

♦ *Mean mark 50%*

COMMENT: Students can read the income levels directly off the graph to save time and then check with the equations given.